

Confusion matrix

Actual class / predicted class	Cancer = yes	Cancer = NO	Total
Cancer = yes	90	210	300 (P)
Cancer = NO	140	9560	9700 (N)
Total	230	9770	10000 (All)
	P	N	

$$① \text{ Accuracy} = \frac{TP + TN}{All} = \frac{90 + 9560}{10000} = \frac{9650}{10000} = 0.965 \times 100 = 96.5\%$$

$$② \text{ error rate} = 1 - \text{accuracy} = 1 - 0.965 = 0.035 = 3.5\%$$

$$③ \text{ Sensitivity} = \frac{\text{True positive}}{P} = \frac{90}{300} = 0.3 = 30\%$$

$$④ \text{ Specificity} = \frac{TN}{N} = \frac{9560}{9700} = 0.986 \times 100 = 98.6\%$$

$$⑤ \text{ precision} = \frac{TP}{TP + FP} = \frac{90}{90 + 140} = 0.391 \times 100 = 39.1\%$$

$$⑥ \text{ Recall} = \frac{TP}{TP + FN} = \frac{90}{90 + 210} = 0.3 \times 100 = 30\%$$

$$⑦ \text{ F1 score} = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} = \frac{2 \times 0.391 \times 0.3}{0.391 + 0.3} = \frac{0.4752}{0.691} = 0.6877$$

$$= 0.688 \times 100 = 68.8\%$$

- ① A Shepherded boy get bored tending the town's flock. To have some fun he cries out, "wolf!" even though no wolf is there. The villages run to protect the flock, but then get really mad when they realize the boy was playing a joke on them. One night, the Shepherded boys see a real wolf approaching the flock and call out "wolf!" The villages refuse to be fooled again and stay in their house. The hungry wolf turn the flock into chops. The town goes hungry, panic ensue. identify (TP, FP, FN, TN)

Sol

Actual \ Predicted	Predicted	
	Wolf = Yes	Wolf = No
Wolf = Yes	warning = correct TP Wolf came = yes	warning Not given FN Wolf came
Wolf = No	FP wolf came = NO	TN No warning No wolf,

- ② Assume there are 100 images, 20 of them depict a cat, the rest do not. A machine learning model, predict the occurrence of cat. 25 of 20 cat image. it also predict absence of a cat in 50 of 70 No cat images identify TP, FP, FN, TN calculate the Precision, Recall, F1 score

Sol

Cat = 20
 Not = 70

Actual \ Predicted	Predicted	
	Cat = Yes	Cat = No
Cat = Yes	TP = 25	FN = 5
Cat = No	FP = 20	TN = 50
	45	55
	100	

$$\textcircled{1} \text{ precision} = \frac{TP}{TP + FP} = \frac{25}{25 + 20} = \frac{25}{45} = 0.555 \times 100 = \boxed{55.6}$$

$$\textcircled{2} \text{ Recall} = \frac{TP}{TP + FN} = \frac{25}{25 + 5} = \frac{25}{30} = 0.833 \times 100 = \boxed{83.3}$$

$$\textcircled{3} \text{ F1 score} = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} = \frac{2 \times 0.555 \times 0.833}{0.555 + 0.833} = \frac{0.92463}{1.388} = 0.666 \times 100 = \boxed{66.7}$$

An email service provider want to classify email as spam or Not spam using features

- Contains "Offers" (yes/no)
- Contains "free" (yes/no)
- Email length (short/long)

sample Dataset

offer free length class

yes yes short spam

NO NO long NO spam

yes yes long spam

NO NO short Not spam

① Calculate Prior probabilities:

- $P(\text{spam})$, $P(\text{Not spam})$

② Computer conditional probabilities for each features

③ using Naive Bayes classify

- (offer = yes, free = No, length = short)

Q1

$$P(\text{spam}) = 3/6 = 1/2$$

$$P(\text{Not spam}) = 3/6 = 1/2$$

Q2) Conditional probabilities

Offer

$$P(O=Y | \text{spam}) = \frac{2}{3}$$

$$P(O=N | \text{spam}) = 1/3$$

Free

$$P(\text{Free} = Y | \text{spam}) = 1$$

$$P(\text{Free} = N | \text{spam}) = 0$$

length

$$P(L=short | \text{spam}) = \frac{2}{3}$$

$$P(L=long | \text{spam}) = 1/3$$

Not spam

$$P(O=Y | \text{Not}) = 1/3$$

$$P(O=N | \text{Not spam}) = 2/3$$

$$P(\text{Free} = Y | \text{Not spam}) = 0/3$$

$$P(\text{Free} = N | \text{Not spam}) = 3/3$$

$$P(\text{length} = Y | \text{Not spam}) = 1/3$$

$$P(\text{length} = N | \text{Not spam}) = 2/3$$

Q3) Using Naive Bayes classify (Offer = yes, Free = No, length = short)

$$P(O=Y, F=N, L=S)$$

Spam

$$\frac{1}{2} \times \frac{2}{3} \times 0 \times \frac{2}{3} = 0$$

$$= \frac{1}{2} \times \frac{2}{3} \times 0 \times \frac{2}{3}$$

$$= P(\text{Spam}) = 0$$

Not spam

$$\frac{1}{2} \times \frac{1}{3} \times 1 \times \frac{1}{3}$$

$$= \frac{1}{18}$$

$$P(\text{Not}) = 1/8$$

$$= 0.055$$

Prediction

$$P(\text{spam}) = 0$$

$$P(\text{Not spam}) = 0.056$$

Prediction will be No spam

ENTROPY

$$H(y) = - \sum_{i=1}^m p_i \log_2(p_i)$$

$$\text{Info}(D) = - \sum_{i=1}^m p_i \log_2(p_i)$$

$$\text{Info}_A(D) = \sum_{j=1}^v \frac{|D_j|}{|D|} \times \text{Info}(D_j)$$

$$\text{Gain}(A) = \text{Info}(D) - \text{Info}_A(D)$$

Assume,

class P: buy-Computer = "Yes"

class N: buy-Computer = "No"

Table

Age	class		I(P,N)
	Yes P	No N	
<= 30	2	3	0.971
31-40	4	0	0
>40	3	2	0.971
	9	5	0.940

$$\begin{aligned}
 I(2,3) &= - \sum p_i \log(p_i) \\
 &= -\frac{2}{5} \log_2\left(\frac{2}{5}\right) - \frac{3}{5} \log_2\left(\frac{3}{5}\right) \\
 &= -\frac{2}{5} (-1.822) - \frac{3}{5} (-0.737) \\
 &= 0.5288 + 0.4422 \\
 &= 0.971
 \end{aligned}$$

$$\begin{aligned}
 I(4,0) &= - \sum p_i \log(p_i) \\
 &= -\frac{4}{4} \log_2(1) - \frac{0}{4} \log_2\left(\frac{0}{4}\right) \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 I(3,2) &= - \sum p_i \log(p_i) \\
 &= -\frac{3}{5} \log_2\left(\frac{3}{5}\right) - \frac{2}{5} \log_2\left(\frac{2}{5}\right) \\
 &= -\frac{3}{5} (-0.737) - \frac{2}{5} (-1.822) \\
 &= 0.4422 + 0.5288 \\
 &= 0.971
 \end{aligned}$$

$$\begin{aligned}
 I(9,5) &= - \sum p_i \log(p_i) \\
 &= -\frac{9}{14} \log_2\left(\frac{9}{14}\right) - \frac{5}{14} \log_2\left(\frac{5}{14}\right) \\
 &= -\frac{9}{14} (-0.6374) - \frac{5}{14} (-1.4854) \\
 &= 0.4097 + 0.524 \\
 &= 0.934
 \end{aligned}$$

$$I(4,2) = -\frac{4}{6} \log_2 \left(\frac{4}{6}\right) - \frac{2}{6} \log_2 \left(\frac{2}{6}\right)$$

$$= -\frac{4}{6} (-0.585) - \frac{2}{6} (-0.585)$$

$$= 0.389 + 0.193$$

$$= 0.582$$

$$I(3,1) = -\frac{3}{4} \log_2 \left(\frac{3}{4}\right) - \frac{1}{4} \log_2 \left(\frac{1}{4}\right)$$

$$= -\frac{3}{4} (-0.415) - \frac{1}{4} (-2)$$

$$= 0.311 + 0.5$$

$$= 0.811$$

$$\text{Info income (D)} = \sum \frac{p_j}{|D|} \times \text{info}(D_j)$$

$$= \frac{4}{14} \times 0.811 + \frac{6}{14} \times 0.582 + \frac{4}{14} \times 1$$

$$= \frac{0.221 + 0.285 + 0.286}{3}$$

$$= 0.264$$

$$\text{Gain income} = 0.940 - 0.264$$

$$= 0.676$$

$$\text{Gain (student)} = 0.151$$

$$\text{Gain (credit-rating)} = 0.048$$

$$Info_{age}(D) = \sum \frac{|D_j|}{|D|} \times info(D_j)$$

$$= \frac{5}{14} \times 0.921 + \frac{4}{14} \times 0.971 + \frac{5}{14} \times 0.971 = 0.694$$

$$= 0.694$$

$$Gain(age) = 0.940 + 0.694$$

$$= 0.246$$

For the attribute income

income	Yes	No	I(P/N)
	P	N	
high	2	2	0.917
medium	4	2	0.917
low	3	1	0.811
	9	5	0.940

$$I(2,2) = - \sum p_i \log_2(p_i)$$

$$= - \frac{2}{4} \log_2\left(\frac{2}{4}\right) - \frac{2}{4} \log_2\left(\frac{2}{4}\right)$$

$$= - \frac{2}{4} (-1) - \frac{2}{4} (-1)$$

$$= \frac{2}{4} + \frac{2}{4} = 1$$

$$= 1$$

Gini index

$$Gini(D) = 1 - \sum_{j=1}^n p_j^2$$

the Gini index based on income

$$Gini(D) = 1 - \sum_{j=1}^n p_j^2$$

$$Gini(D) = 1 - \left[\left(\frac{9}{14} \right)^2 + \left(\frac{5}{14} \right)^2 \right] \left[\begin{array}{cc} L & -4 \\ M & -6 \\ A & -4 \end{array} \right] \left. \begin{array}{l} -10 \\ \frac{4}{14} \end{array} \right\}$$

$$= 1 - [0.413 + 0.127]$$

$$= 1 - 0.54$$

$$= 0.46$$

Partition of data set with respect to income

$$D_1 : (\text{low, medium}) - 10$$

$$D_2 : (\text{high}) - 4$$

$$Gini_{(\text{income})}(D) = \frac{10}{14} Gini(D_1) + \frac{4}{14} Gini(D_2)$$

$$= \frac{10}{14} \left[1 - \left[\left(\frac{4}{10} \right)^2 + \left(\frac{3}{10} \right)^2 \right] \right] + \frac{4}{14} \left[1 - \left[\left(\frac{2}{4} \right)^2 + \left(\frac{2}{4} \right)^2 \right] \right]$$

$$= \frac{10}{14} [1 - [0.16 + 0.09]] + \frac{4}{14} [1 - [0.25 + 0.25]]$$

$$= \frac{10}{14} [1 - 0.25] + \frac{4}{14} [1 - 0.50]$$

$$= \frac{10}{14} [0.75] + \frac{4}{14} [0.50]$$

$$\text{Gain ratio} = \frac{\text{gain}(A)}{\text{Split info}(A)}$$

$$\text{Split info}(D) = - \sum_{j=1}^N \frac{|D_j|}{|D|} \times \log_2 \frac{|D_j|}{|D|}$$

$$= \text{Split info}_{(\text{age})} = - \sum \frac{|D_j|}{|D|}$$

$$= -\frac{5}{14} \log_2 \frac{5}{14} - \frac{4}{14} \log_2 \left(\frac{4}{14}\right) - \frac{5}{14} \log_2 \left(\frac{5}{14}\right)$$

$$= -\frac{5}{14} (-1.4854) - \frac{4}{14} (-1.8074) - \frac{5}{14} (-1.4854)$$

$$= 1.58$$

$$G_R = \frac{\text{Gain}}{\text{Split info}} = \frac{0.246}{1.58} = 0.156$$

$$\Rightarrow \text{Split info (income)} = - \sum \frac{|D_j|}{|D|}$$

$$= -\frac{4}{14} \log_2 \frac{4}{14} + \frac{6}{14} \log_2 \frac{6}{14} + \frac{4}{14} \log_2 \frac{4}{14}$$

$$= -\frac{4}{14} (-1.8074) - \frac{6}{14} (-1.2224) - \frac{4}{14} (-1.8074)$$

$$= 1.557$$

$$G_R = \frac{\text{Gain}}{\text{Split info}} = \frac{0.029}{1.557} = 0.019$$

$$= 0.443$$

$\Rightarrow D_1 \{ \text{medium, High} \}$

$D_2 \{ \text{low} \}$

$$= 10/6/4$$

$$\frac{4}{3/1}$$

$$Gini(\text{income}) (m/H) = \frac{10}{14} \left[1 - \left[\left(\frac{6}{10} \right)^2 + \left(\frac{4}{10} \right)^2 \right] \right] + \frac{4}{14} \left[1 - \left(\frac{3}{4} \right)^2 + \left(\frac{1}{4} \right)^2 \right]$$

$$= \frac{10}{14} \left[1 - [0.36 + 0.16] \right] + \frac{4}{14} \left[1 - 0.5625 + 0.0625 \right]$$

$$= \frac{10}{14} [0.48] + \frac{4}{14} [0.37]$$

$$= \frac{4.8}{14} + \frac{1.48}{14}$$

$$= \frac{6.28}{14}$$

$$= 0.449$$

$D_1 \{ \text{low, High} \}$

$D_2 \{ \text{medium} \}$

$8 \rightarrow 5, 3$

$6 \rightarrow 4, 2$

$(Gini = 0.458)$
(~~low~~)

Gini index

D_1	D_2	Value
HH	H	0.443
WH	M	0.458
MH	L	0.450

Handwritten signature
11/8/2